

Information Systems Development Methods and Tools Course Lecture Notes and AI Usage Guidelines

Experiment 1: Evaluating Existing Information Systems

Experimental Purpose

- Effectively summarize the advantages and disadvantages of common information systems/applications through observation and analysis.
- Understand and verify the evaluation methods of information systems.
- Effectively apply this experience to one's own system development project.

Experimental Procedure

Grouping

Form groups of 1-3 people freely and elect one group leader.

Material Reading

Read this material and earnestly study the relevant knowledge about pain points, pleasure points, and itch points in product development.

[📖 Pain points, itch points, and pleasure points in product R&D](#)

Evaluate the system based on these three dimensions. You can follow the definitions in the link text or search for relevant definitions on your own. If you use definitions from other sources, you need to indicate the source in the report.

Select System

Select a common system/website/application (at least most of the group members have used it) as a sample for analysis. Explain the reasons for the selection in the report.

Brainstorming

The team leader is responsible for organizing everyone to brainstorm, pool ideas, and sort out the strengths and weaknesses of the system.

For the identified shortcomings, find ways to propose improvement countermeasures.

Write a report

As a group, write up the results of the brainstorming in a text-and-image format. The report should include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. However **when submitting, please export it as a PDF file** and submit it via Base. The approximate format structure is shown here. [📄 Experiment Report](#)

Record Video

Present the content of the manuscript through short videos.

Please express yourself naturally and don't read from a script. Because both teachers and students **can read**.

Here are two reference videos for you.

The first case is the cloud storage provider Dropbox. When they were starting their business back then, they hadn't yet developed a product prototype. So they created this animation and showed it to investors, successfully securing investment.

[what_is_dropbox.mp4](#)

The second video was made by the instructor (Mr. Wang Shuyi) when he participated in the Innovation Design Workshop in June 2017. From the creative discussion to the completion of recording, this video only took 15 minutes. It's not about how high the quality is; the main purpose is to give you a boost - even if it's so rudimentary, there's no need to feel embarrassed.

[Blockchain Tracking Product Concept Showcase.mp4](#)

Video Release

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For an introduction to asparagus, you can [click here](#). (Video time limit: two minutes)

Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for the teaching assistant to conduct the review.
- **Before 11:15**, each group shall complete the interactive evaluation. Evaluate the other group's experimental report and video, and raise targeted questions to a specific classmate based on the specific division of labor
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Evaluation Dimensions of Computer-Based Tasks

- Complete content. It means that all tasks in the experimental requirements have been completed without omission.
- Submitted on time. It refers to completing the specified operations before the deadline, including material submission and evaluation, etc.
- Maintain a proper attitude. Avoid misconduct such as cheating, plagiarism, and careless and perfunctory work.

Precautions

- Since there are many steps involved, please make sure to hurry up. Remember that "completion is more important than perfection". Do not overdo it by constantly making revisions, which may cause you to miss the submission deadline. Late submissions will be handled in accordance with the rules for late assignments.
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Experiment 2: Outlook on System Development Goals

Experimental Purpose

Envision the functional vision of the system and present it to your potential users using appropriate methods.

Experiment Introduction

After summarizing the advantages and disadvantages of existing systems in Experiment 1, you should start **setting goals for the development of your own system** . The most important aspects of this are **positioning, boundaries, and iteration** .

The so - called **Positioning** is to identify what pain points your system aims to address. These pain points should be **real** .

The so - called **boundary** refers to what your system can and cannot do. No product can be a "cure - all". Understanding what the system **cannot do** will give you a clearer understanding of your product and help you avoid going astray. Here, you may refer to [this explanation on the help page of the note - taking tool flomo](#) .

flomo 不擅长什么

我们深知做好一件小事就已经很不容易，所以不会试图在功能层面 All in one，如果你对 flomo 有下面的预期，那么可以选择我们推荐的其他产品替代。

- 不擅长写 文档撰写，推荐[石墨文档](#)
- 不擅长做 文档收藏，推荐[印象笔记](#)
- 不擅长做 TODO，推荐[滴答清单](#)
- 不擅长做 思维导图，推荐 [XMind](#)
- 不擅长做 助记卡片，推荐 [Anki](#)
- 不擅长做 电子表格，推荐 [Airtable](#)

另外我们很敬佩 [Notion](#) 、 [RoamResearch](#) ，因为他们给了 flomo 很多启发。

The so - called **iteration** means that when you first envision a system, you should have a vision to provide a goal for the subsequent development process. However, you don 't need (and it's basically impossible) to make it perfect right from the start. You will have plenty of opportunities **to adjust and improve** .

Experimental Procedure

Brainstorming

The team leader is responsible for organizing everyone to use [mind maps](#) , [whiteboards](#) and other forms to conduct brainstorming, pool ideas, and initially form **at least 2** (the more the better) different system development goals. For each goal, explain the following matters:

Who are the Target Users/Customers?

What pain points are being addressed? (Note that **problems that have already been actually solved by existing systems cannot be regarded as pain points** . For example, you cannot **pretend that the QR code scanning function does not exist** and then design one yourself.)

What features are provided?

Make a choice

Within the group, compare the results of the above brainstorming. Rank at least two different system development goals. Indicate which one is more **preferred** and **why**.

Write a report

As a group, write up the results of the brainstorming in a text-and-image format. The report should include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. However, **when submitting, please export it as a PDF file** and upload it publicly via the Feishu group chat, with the teaching assistant responsible for backup. The approximate format structure is shown here. [📄 Experiment Report](#)

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- Conduct experiments within the scope of legality and compliance to ensure personal safety

Experiment 3 Interview Target User

Experimental Purpose

Determine whether the topic is appropriate by conducting interviews with Target Users, and further collect users' pain points and needs.

Experiment Introduction

In [Experiment 2](#) , you have already attempted to formulate more than two system development goals and prioritized them. In addition to having a good vision for the goals, you also need to further understand **the real needs of users** . This is an important prerequisite for conducting feasibility analysis.

If the topic selection is appropriate, the next step of work can be carried out based on the feedback; if not, it doesn't matter. After class, search for a new topic again and contact suitable new users. However, note that if the operation of this experiment is not rigorous and the topic selection does not meet the needs of the target user, there may be serious problems where the project **cannot progress and be delivered smoothly** . Please **take this experiment seriously and solemnly** .

Experimental Procedure

Advance Preparation

Select appropriate users. [Experiment 2](#) has been pre-arranged, and users need to be found according to your system objectives. Note that **users cannot be classmates participating in this course, nor can they be direct relatives of the developers**. For each system objective, the number of users is 1-2.

Sort out the user interview outline.

Prepare the tools for the interview. It is recommended to use Feishu Meeting. The main reason is that it facilitates recording the interview process and can generate a Feishu Minutes document with one click. The document includes automatically transcribed text, which is helpful for subsequent organization and analysis.

Prepare the content for presentation. You can reuse [the materials generated from Experiment 2](#) , but be sure to adjust them according to the user's actual situation for better communication.

User Interview

You need to at least complete the following steps:

Opening remarks. Conduct necessary communication and small talk to create a suitable interview atmosphere, and explain the purpose of this interview.

Information collection . This includes the user's current work/operation process, their feelings during this process, especially those real pain points.

Describe/Demonstrate Your System (Prototype). You can use methods such as screen sharing to showcase the expected functions of your system. Since your system does not yet exist, you can use hand-drawn interfaces and other means to give users a more intuitive experience.

Feedback on system objectives. Are users interested in the system you described/demonstrated? Which aspects meet their needs? Which aspects deviate? And so on. Feedback must be detailed, and any unclear points need to be communicated and confirmed

with users. Otherwise, misunderstandings like "you talk about the front gate tower, but the user hears the hip joint" are likely to occur, which is not conducive to the subsequent work.

Cost Affordability Survey. Ask users how much cost they are willing to pay for the system. Note that cost includes not only economic costs, such as subscription fees/one-time purchases, but also time costs, learning costs, etc. associated with use.

Follow - up willingness . In the subsequent system development phase, you still need users to provide continuous feedback. Do users have the time and willingness to offer assistance?

Result Analysis

Based on user feedback, analyze the answers to the following questions:

Does the anticipated user pain point actually exist?

Can the solution (system) you provided address the pain points of users?

Are users willing to pay the cost for your system?

After comprehensive consideration, the answer to this question is provided as follows:

Is the system development objective you provided in [Experiment 2](#) appropriate?

Write a report

As a group, write up the above analysis results in a document. It is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork.**When submitting, please export it as a PDF file, and upload it via the Feishu group form (e9c9e17fy.feishu.cn).**

Please attach the transcript of the user interview to the end of the report. Note that if you recorded the interview video using Feishu Meeting, the system will automatically generate a text version via Feishu Minutes. You only need to attach the link to the Feishu Minutes document (check the permissions to ensure others can open the link normally), and there is no need to paste the full text. This is why we recommend everyone to use Feishu for user interviews, as it is convenient and fast.

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

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Reference Materials

[Here](#) is a learning material about user interviews for everyone's reference. Since the content of the material is relatively extensive, here is a relatively simple operational prompt, and you can [refer to this document](#).

Experiment 4 System Planning

Experimental Purpose

1. Proficiently master the relevant graphic production techniques in the system planning phase;
2. Conduct a feasibility analysis;
3. Determine the topic for the development system this semester

Experiment Introduction

In previous experiments, we analyzed the development goals of different systems, conducted interviews with target users, and determined the development goals. Starting from this experiment, we will begin to act in accordance with this goal.

Let's review the system development lifecycle here. Note that this is an iterative diagram.



In this diagram, the first part is the system's planning (Planning).

The content mainly includes: (1) Drawing of PERT charts and Gantt charts. (2) Project feasibility analysis

Experimental Procedure

Set the progress of subsequent stages

Please ask team members to fully discuss, set the duration for the subsequent project development phase, and make good division of labor.

Additionally, if there are deficiencies in the corresponding basic knowledge and skills of system development (such as planning to use a new framework or development language), sufficient time also needs to be allocated for learning and training.

It is recommended that when setting the deadline for each stage **leave some buffer (be flexible)** , and in actual operation **try to complete it ahead of schedule** .

Remember? "Done is better than perfect." Build a prototype earlier, and then iterate "leisurely". Progress is **in your own hands**, and your mood will turn pleasant. Never leave too much time in the early stages while lacking flexibility in the subsequent stages. Even more so, don't procrastinate until the last minute to start working on completion. Uncontrollable factors may arise at any time, and if there is not enough flexibility, it may lead to the project being unable to be completed and the final assignment being unable to be submitted on time.

Task Setting

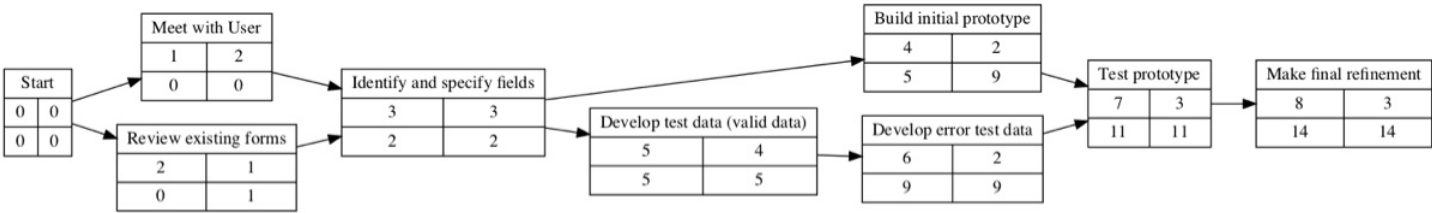
Please combine the actual situation, sort out the specific tasks that need to be completed, and list them in a table like the following.

任务编号	任务描述	任务时长	前导任务编号
0	开始	0	--
1	会见用户	2	0
2	回顾现有表格	1	0
3	确认与选择领域	3	1, 2
4	构造初始类型	2	3
5	研发测试数据	4	3
6	研发错误测试数据	2	5
7	测试类型	3	4, 6
8	做最终修订	3	7

Of course, your tasks should be more detailed than a table like this. In terms of time units, use "**days**" as the smallest time unit.

Draw a PERT Chart

According to the content of the lecture, present the detailed table as a PERT chart.



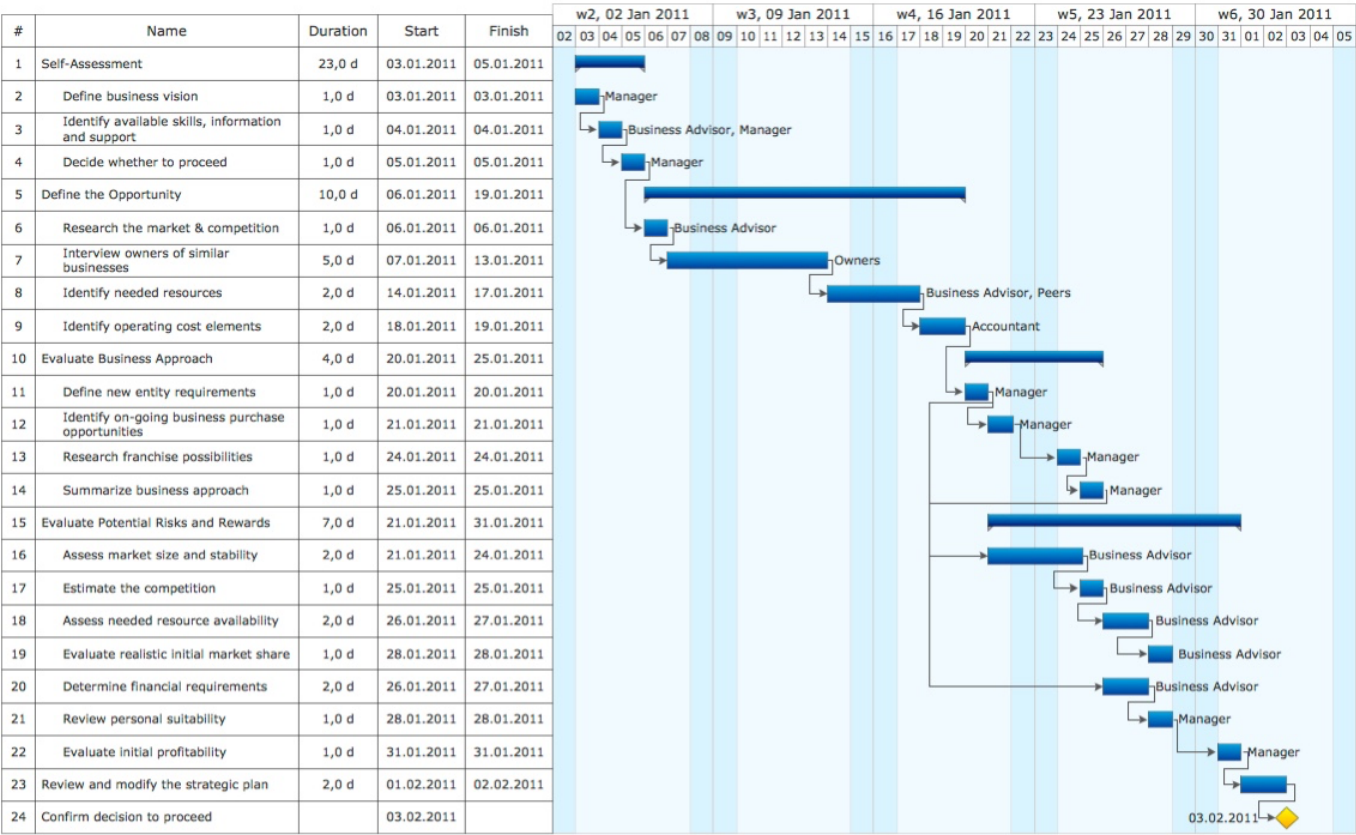
Next, don't forget to find the node number sequence of the Critical Path.

It is recommended to use the online collaboration tool Feishu Docs to complete the drawing of the above table. Here is the link to the demonstration video: [Drawing a PERT Chart with Feishu Docs](#)

Draw Gantt Chart

To clarify the dependencies of tasks in the project and streamline each person's work objectives, it is necessary to draw a **Gantt Chart**.

Strategic Plan for New Business



An example of creating a Gantt Chart in Feishu Docs (Base) has already been demonstrated in class, so it will not be repeated here.

Note that all drawings must be computer-generated, and hand-drawn ones are not allowed.

Task Time Estimation

Present the task time allocation mentioned in the PERT Chart and Gantt Chart with specific details.

First, list the time required for each task and the total duration. Then, provide detailed justifications for the various time schedules given above. Do not subjectively determine the duration of any task, and avoid logical contradictions and unreasonable task arrangements. There must be a basis to indicate the time required for each task. For example, tasks that require technical preparation, learning, or review should have their time taken into account.

Additionally, the task progress measured in weeks or months should be mathematically calculated (in the form of a mathematical expression), ultimately providing the weekly or monthly task duration as well as the total duration of the task (ultimately expressed in days). Note that the mathematical calculation formula should be written out and accompanied by detailed textual explanations.

The minimum unit of measurement for all the above task durations is days.

Project Feasibility Analysis

Analyze the project feasibility based on the results of leading experiments such as brainstorming and user interviews. The analysis should be based on facts.

Economic Feasibility

Economic feasibility mainly assesses the development costs of a project and the potential economic benefits that may be obtained after the project's success. Only when the development costs can be controlled within the acceptable range of the enterprise is the project likely to be approved for implementation. The scope of economic benefits is widely considered, including direct cash income, expected sales revenue, technology accumulation, business support, market development, improvement of customer satisfaction, etc.

Technical Feasibility

The functions and performance to be achieved by the Technical Feasibility Assessment System, as well as the technical capability constraints, are mainly demonstrated through the following questions:

- Is the Prior Art and the current state of IT technology development sufficient to support the achievement of system objectives?
- Are the existing resources (technical personnel, technical accumulation, component library, software and hardware conditions, etc.) sufficient to support the project implementation?
- Among the set system objectives, which ones may encounter strong technical barriers?

Legal Feasibility

The development of the legal feasibility assessment system may give rise to potential infringement or legal liability, including contract formation and terms, responsibility assignment, infringement situations, breach of contract, and dispute resolution. It also includes restrictions imposed by national policies and laws, such as the use of unrecognized encryption algorithms or unauthorized use of protected software technologies.

Implementation Feasibility

, also known as operational feasibility, mainly assesses the extent to which the expected software system can be applied in a real environment and the obstacles during the implementation process. It includes various impacts on users:

- Impact on existing IT facilities
- Impact on the user's organization
- Impact on existing business processes
- Impact on Location
- Impact on funding expenditure

Determine the topic

Once the topic is determined, if there are subsequent changes, they must be implemented in accordance with the rules. Please be sure to exercise caution. After the topic is determined here, the team leader should sign by hand on the paper version of the experimental report.

If the feasibility analysis fails, a negative result report can be submitted, and a planning report including the finally determined topic selection should be supplemented before the next machine operation.

Write a report

Write the experimental report in groups using text and images. Note that in the reports of each stage, not only should the corresponding characteristic charts be included, but also **there must be matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, for the sake of fairness, please export it as a PDF file and upload it via the Feishu group form (abc9e17fy.feishu.cn).**

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Experiment 5 System Analysis

Experimental Purpose

Learn and create user profiles and data flow diagrams.

Begin to write the system analysis report.

Experiment Introduction

According to the system development lifecycle, this experiment will commence the second part of system development - system analysis. The main content of this hands-on experiment is **user profiling** and **data flow diagram** drawing.

Experimental Procedure

Requirement Analysis

Identify the **end users** of the system and analyze the relevant requirements.

When drawing a data flow diagram, it is necessary to consider what functions the system as a whole can perform, what relationships it has with the external world, and what the most important input and output data are. If the system functions are complex, it is necessary to consider whether they can be decomposed into independent sub-functions.

Please consider the above questions based on the actual situation of your system.

User Portrait

Learn [user portrait technology through learning materials](#).

Please further think and dig into user needs based on the completed user interviews, and comprehensively process the collected user characteristics (identity/occupation/behavior/lifestyle/gender, etc.) and key user behavior data. Note that if the information collected from user interviews is incomplete, you can find **the same users** to supplement the collection of relevant information.

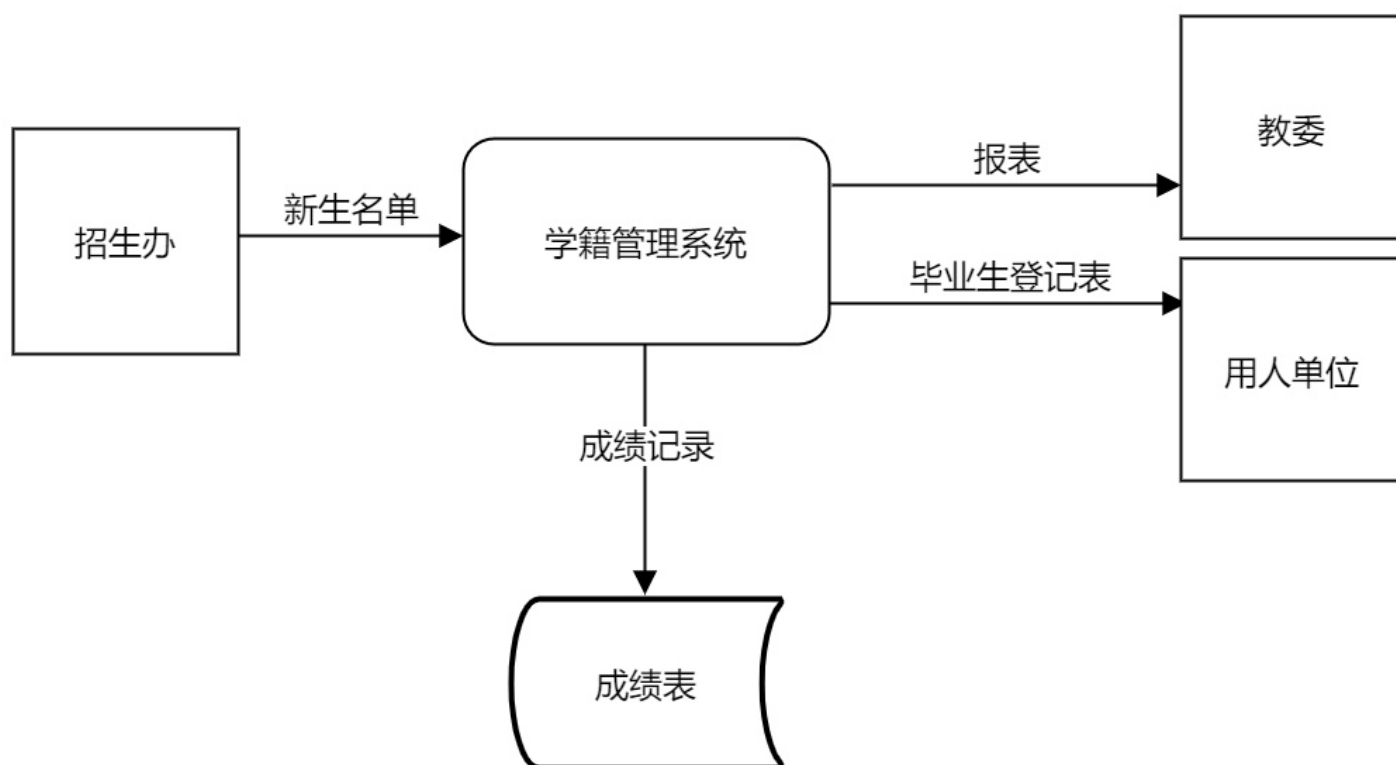
Due to time constraints, a maximum of three user profile images can be submitted for this experiment. For the remaining images, please complete them after class and include them in the subsequent submission of the "System Analysis Report".

It is recommended to use [BeeArt](#) online collaboration tool to complete the above user portrait diagram. [Here is a simple tutorial on using BeeArt to draw user portraits.](#)

Draw a data flow diagram

Based on the content obtained from the requirements analysis, the functions (events/processes) involved in the system are represented through data flow diagrams.

Example:



A system, called a "system", must be composed of **several basic functions** organically combined. The drawing of a data flow diagram must include **overall** and **detailed decomposition** diagrams. Due to time constraints, this experiment only requires the submission of 1 overall data flow diagram and at least 2 decomposed data flow diagrams. The remaining diagrams (decomposed data flow diagrams corresponding to the remaining functions) should be supplemented after class and reflected in the "System Analysis Report" submitted later.

It is recommended to use the online collaboration tool Feishu Docs to complete the drawing of the above data flow diagram. Here is the link to the tutorial: [Feishu Data Flow Diagram](#)

Write a report

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See here for the general format structure. [Basic style of the Information System Practical Report](#)

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- Conduct experiments within the scope of legality and compliance to ensure personal safety

Experiment 6 System Analysis

Experimental Purpose

Learn and create **Use Case Diagrams** and **UML Class Diagrams**, and continue to write the system analysis report.

Experiment Introduction

This experiment continues with the second part of system development - system analysis. Add **use case diagrams** and **UML class diagrams**, to prepare for the final completion of a comprehensive system analysis report.

Experimental Procedure

Requirement Analysis

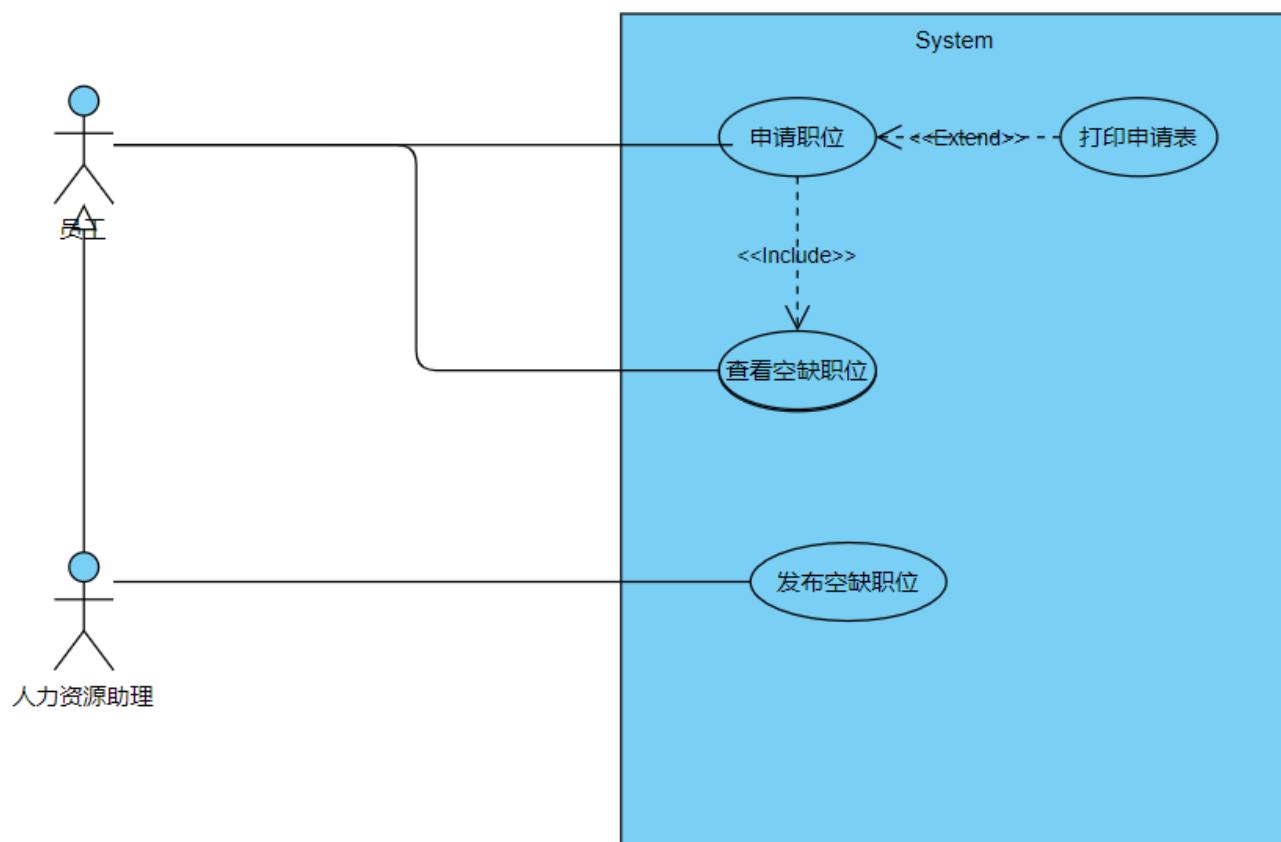
When drawing a use case diagram, in addition to clarifying who the end users of the system are, it is also necessary to consider what functions the system can **provide** and **the purpose of users using the system**.

To draw a UML class diagram, one must clearly understand the **various business objects within the system domain and their relationships**, thereby helping users better understand the business domain in which the system operates.

Please consider the above questions based on the actual situation of your system and **form written records**.

Use Case Diagram

Based on the content obtained from the requirements analysis, the users/participants and functions involved in the system are presented through use case diagrams.



Here is a basic introduction to use case diagrams:[UML Modeling - Use Case Diagram \(Use Case Diagram\) - lcl - Cnblogs](#)

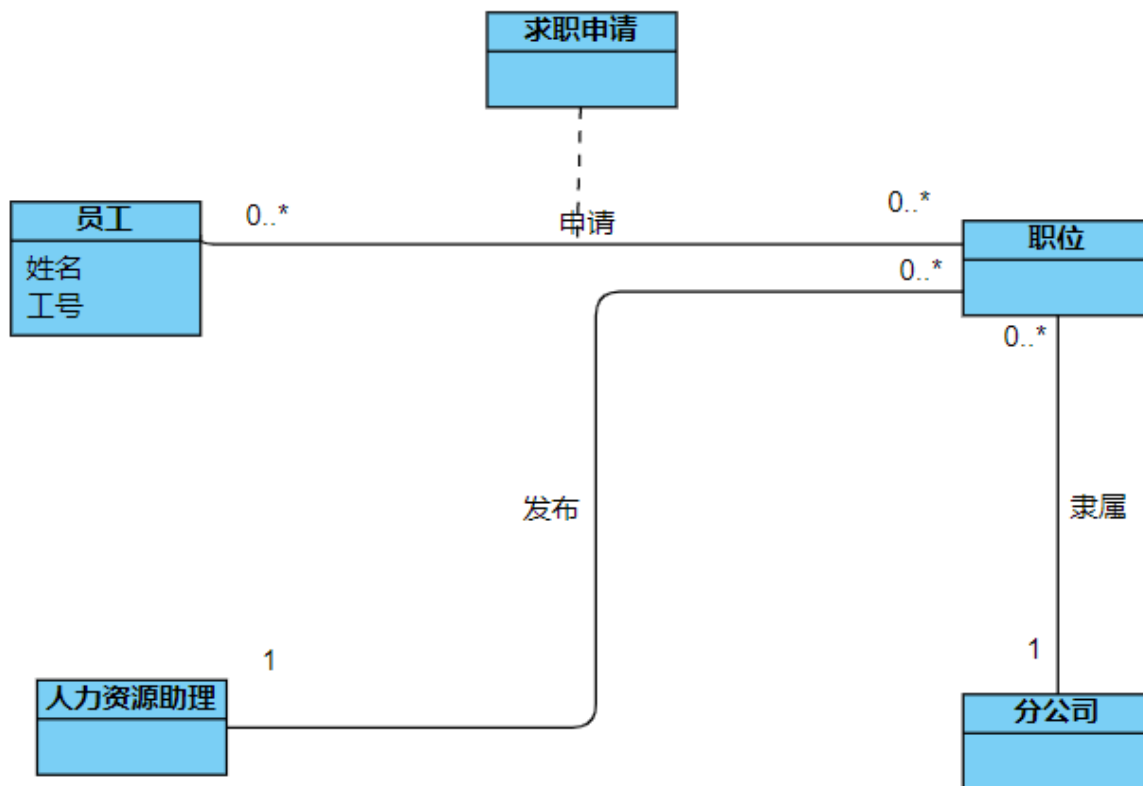
Here is a reference video tutorial for drawing use case diagrams:[Learn UML Use Case Diagrams in 6 Minutes_Bilibili](#)

It is recommended to use [Process On](#) to complete the drawing of the above use case diagram.

Here is the [User Manual](#).

UML Class Diagram

By analyzing each business object within the system domain and the relationships between them, the drawing of the UML class diagram is completed.



Here is a basic introduction to UML class diagrams:[Understand UML Class Diagrams in Five Minutes - shindoyang - Cnblogs](#)

Here is a reference video tutorial for drawing UML class diagrams:[Learn UML Class Diagrams in 6 Minutes_Bilibili](#)

It is recommended to use [Process Onto](#) to complete the drawing of the above use case diagram.

Here is the [User Manual](#).

Note that for the drawing of the above two types of diagrams, each development team first needs to determine the number of diagrams to be drawn respectively (i.e., judge how many use case diagrams and UML class diagrams need to be drawn respectively based on the actual situation of the system). Secondly, considering the limited computer lab time, each development team is required to **complete the drawing of two diagrams for each type respectively**. Finally, if the number of diagrams to be drawn exceeds two, the drawing of the remaining diagrams can be completed during the computer lab session when the overall system analysis report is submitted.

(Note that the report should include a recheck of implementation feasibility and necessity. If there are any issues, it is recommended to make timely adjustments.)

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and upload it publicly via ebc9e17yfy.feishu.cn. **The teaching assistant is responsible for backup**.

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

Since the free version of Feishu has limited space, please try to **avoid directly uploading video files** to the group to prevent space limitations from affecting other students' usage. Please fill in the sharing link at (ebc9e17yfy.feishu.cn). It is recommended to use [Asen](#) screen recording app to record and automatically publish videos.

Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年4月2日 实验六](#)).
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

Precautions

- Since there are many steps involved, please make sure to hurry up. Remember that "completion is more important than perfection". Do not overdo it by constantly making revisions, which may cause you to miss the submission deadline. Late submissions will be handled in accordance with the rules for late assignments.

- Team leaders should do a good job in division of labor and collaboration, fully mobilize the enthusiasm of team members, ensure timely completion, and guarantee the quality of the content.
- To protect privacy, the scope of publication for documents and videos **is restricted to the Feishu group of this course**, please **do not publicly disclose** or disseminate them.
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- The use of AI assistants is encouraged and promoted in this course, so you don't need to have any concerns. However, you must explain what issues or areas for improvement there are in the AI-generated results, and how you further communicated based on the preliminary answers provided by the AI. **Submitting AI-generated answers directly without review or modification is considered plagiarism** and will result in point deductions
- Conduct experiments within the scope of legality and compliance to ensure personal safety
- The recommended software (tools) are suggestions and not mandatory requirements

Experiment 7 System Analysis

Experimental Purpose

Write system analysis reports and proficiently master relevant graphics production techniques.

Experiment Introduction

The content of this hands-on experiment mainly includes group logo design and empathy map drawing, as well as reviewing and analyzing the content of the previous report.

Experimental Procedure

Requirement Analysis

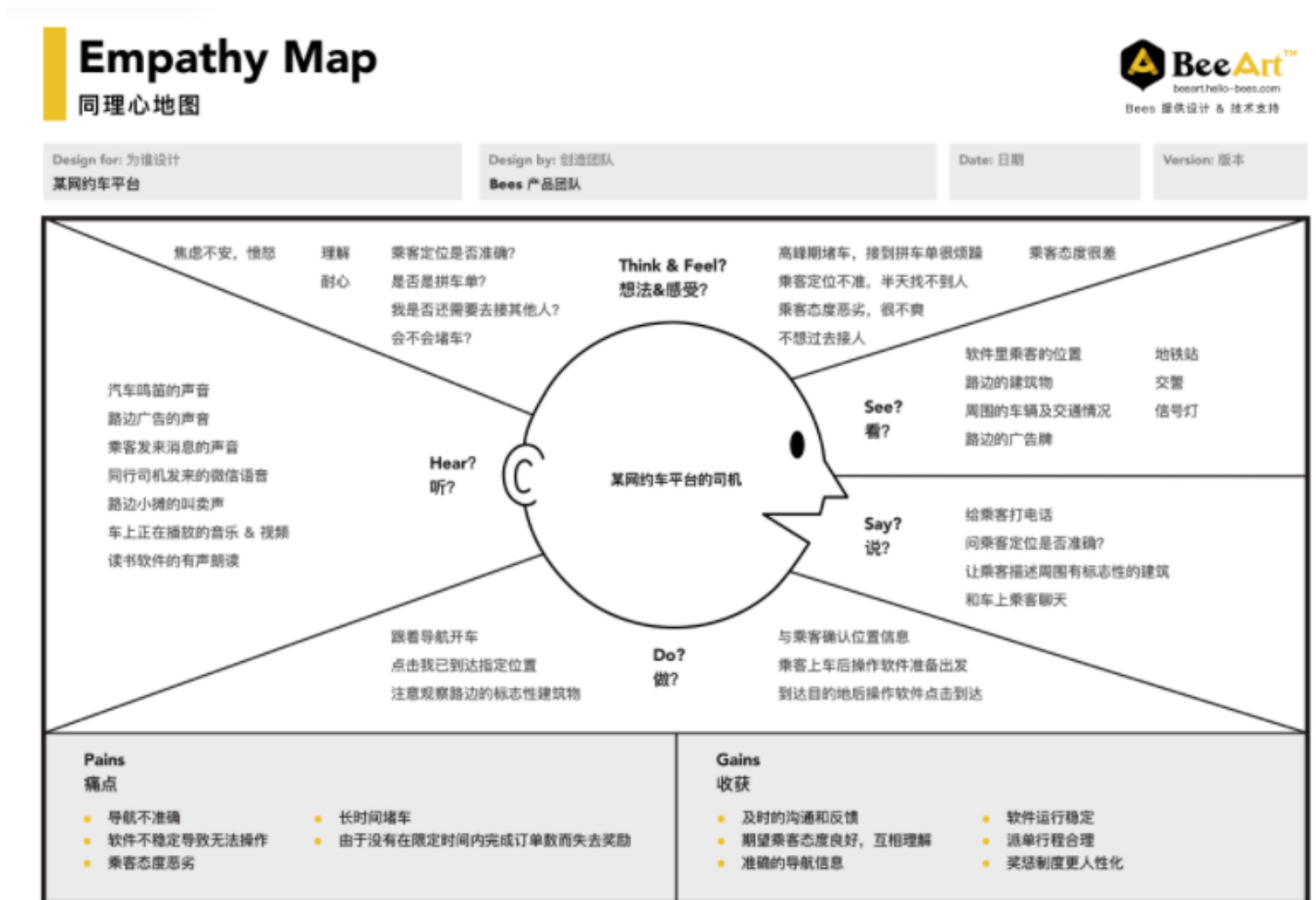
Regarding the creation of empathy maps, first, it is necessary to consider which users have problems (Who). Second, it is necessary to gain in-depth understanding of what these users' thoughts, hobbies, and pain points are respectively (What). Finally, think about what makes them happy and satisfied (How).

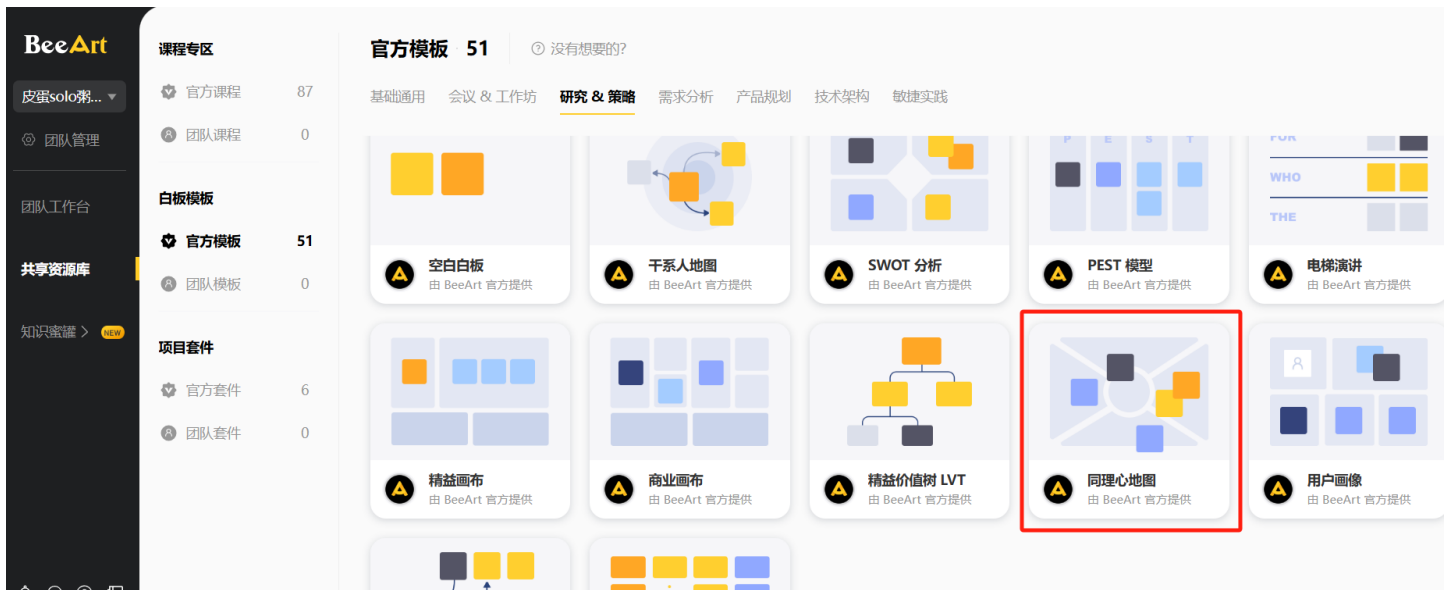
Empathy Map

Empathy maps can help system developers more clearly analyze the issues that users are most concerned about, thereby finding better solutions to problems.

Please cluster the collected key user behavior data based on the results of user interviews from previous experimental processes, think about several questions in the "Needs Analysis", and use the drawing of empathy maps to find more accurate entry points, so as to solve the corresponding problems.

The drawing of the empathy map can refer to the first half of this [video](#).





It is recommended to use BeeArt online collaboration tool (<https://www.beeart.com>) to complete the drawing of the above empathy map. Here is the tool introduction video (<mp.weixin.qq.com>).

Regarding the drawing of empathy maps, each development team first needs to determine the number of maps to be drawn (i.e., judge which users will have problems based on the actual situation of the system). Second, considering the limited computer time, each development team is required to draw at least 2 empathy maps. Finally, if the number to be drawn exceeds two, the drawing of the remaining maps can be completed during the subsequent class session when the overall system analysis report is submitted.

Note that after the empathy map is completed, it is necessary to review and analyze the content of the previous report again, summarize whether there are any issues that were not discovered at the time, and list them in tabular form in this report.

Logo

Each group needs to pool their wisdom and efforts to design a logo for their group's system. The following are reference examples.

Information Management Books (IMOK)



Everyone can complete the design through photo synthesis of real people, or by means of drawing, PS, AI generation, etc., **and finally, the design inspiration of the logo should be elaborated in this analysis report.**

Additionally, please act within your capabilities and pay attention to safety during the photo-taking process.

Content Filling

In previous experiments, some content (drawing of relevant graphics) was not required to be fully presented due to limited computer time. In this experiment, you are required to complete these contents, including the drawing of graphics for all functional designs of the system, such as use case diagrams, class diagrams, data flow diagrams, etc. **There must be corresponding written descriptions.**

Important Announcement

The next experiment will be the last one in the system analysis phase. That is to say, each development team will need to report to the Target Users on the progress of this phase and collect user feedback. Therefore, we need everyone to **contact in advance**.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions** .

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group form (ebc9e17yfy.feishu.cn) , **with the teaching assistant responsible for backup** .

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

Because the free version of Feishu has limited space, please try **not to upload video files directly** to the group, so as not to affect other students from using the upload to document function due to space limitations. Please upload links through the form (ebc9e17yfy.feishu.cn). It is recommended to use the [asparagus](#) screen recording application to record and automatically publish the video.

Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年4月9日 实验七](#)) .
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

Precautions

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Experiment 8 System Analysis

Experimental Purpose

Write the overall system analysis report and present it to the Target User.

Experiment Introduction

This hands-on session is the last experiment in the system analysis phase. Each development team needs to modify and improve the issues listed in the previous experiment, fill in relevant content as required, and finally submit a complete system analysis report, present it to the Target User, and then collect detailed feedback.

Experimental Procedure

Problem Correction

In the previous experiment, we re-examined and analyzed the previous system analysis reports and identified some issues that needed to be modified and improved. In this experiment, we need to make practical modifications to these issues respectively.

Note that the list of issues obtained last time needs to be placed at the beginning of the Report for Experiment 8. Also, indicate the areas that need to be revised, and explain how to make the revisions and the reasons for them in the form of text and images.

Overall Review (General Report of the System Analysis Phase)

Write a comprehensive system analysis report. Note that writing the overall report is not simply aggregating the previous experiments, but requires systematically and logically connecting the several experimental reports from the system analysis phase, and organizing and summarizing the supplementary graphics into the overall report. The selection of excellent groups will also assess whether the overall report is logical (rather than a simple aggregation).

Note: If you complete the system analysis report by 9:00, you can sign up with the teaching assistant to get the opportunity to use AI Agent-related products (Manus, Genspark, Flowith) to develop a prototype system.

Phase Report (User Further Feedback Report - Phase Report on the System Analysis of Experiment VIII)

After the overall system analysis report is completed, it is necessary to present and explain the report content to the target users to further obtain user feedback. It is required to attach screenshots, text content, video recordings of the feedback process, user feedback issues, and the provided reflections and solutions to this experimental report. Pay attention to the minimum word count (500 words) for the user communication and feedback process to avoid overly brief communication content.

In the reporting video with the Target User, users do not need to show their faces. Moreover, to save Feishu group memory, it is still recommended that everyone use link methods such as Lusun to attach the video link to the report. Meanwhile, if you record an interview video using Feishu Meeting, the system will automatically generate a text version through Feishu Minutes, and you only need to attach the link to the Feishu Minutes document (check the permissions to ensure others can open the link normally), without pasting the full text. This is why it is recommended that everyone use Feishu for user interviews, as it is convenient and fast.

Iteration

Guided by the principle that **completion is more important than perfection**, we hope that everyone will not be overly nitpicky when writing reports and will pay attention to submitting them on time. However, please note that if there is more spare time after completing the report, do not simply be content with completion; you can continue to iterate to make the completed work more perfect.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

Today's reports are divided into two parts. **One is the Report on Experiment 8, and the other is the General Report on the System Analysis Phase.**

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group form (ebc9e17yfy.feishu.cn) , **with the teaching assistant responsible for backup** .

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release (Two Videos, One Interview Video and One Explanation Video)

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Time Node

- The experiment report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年4月16日 实验八](#)) .
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
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Precautions

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Experiment 9 System Overall Design

Experimental Purpose

Complete the overall system design report for this experiment by learning to draw structure diagrams and design software classes.

Experiment Introduction

According to the system development lifecycle, we have completed the tasks of the second part, system analysis.

This experiment will initiate the third part of system development - system design. The main content of this hands-on experiment includes the drawing of structure diagrams and the design of software classes.

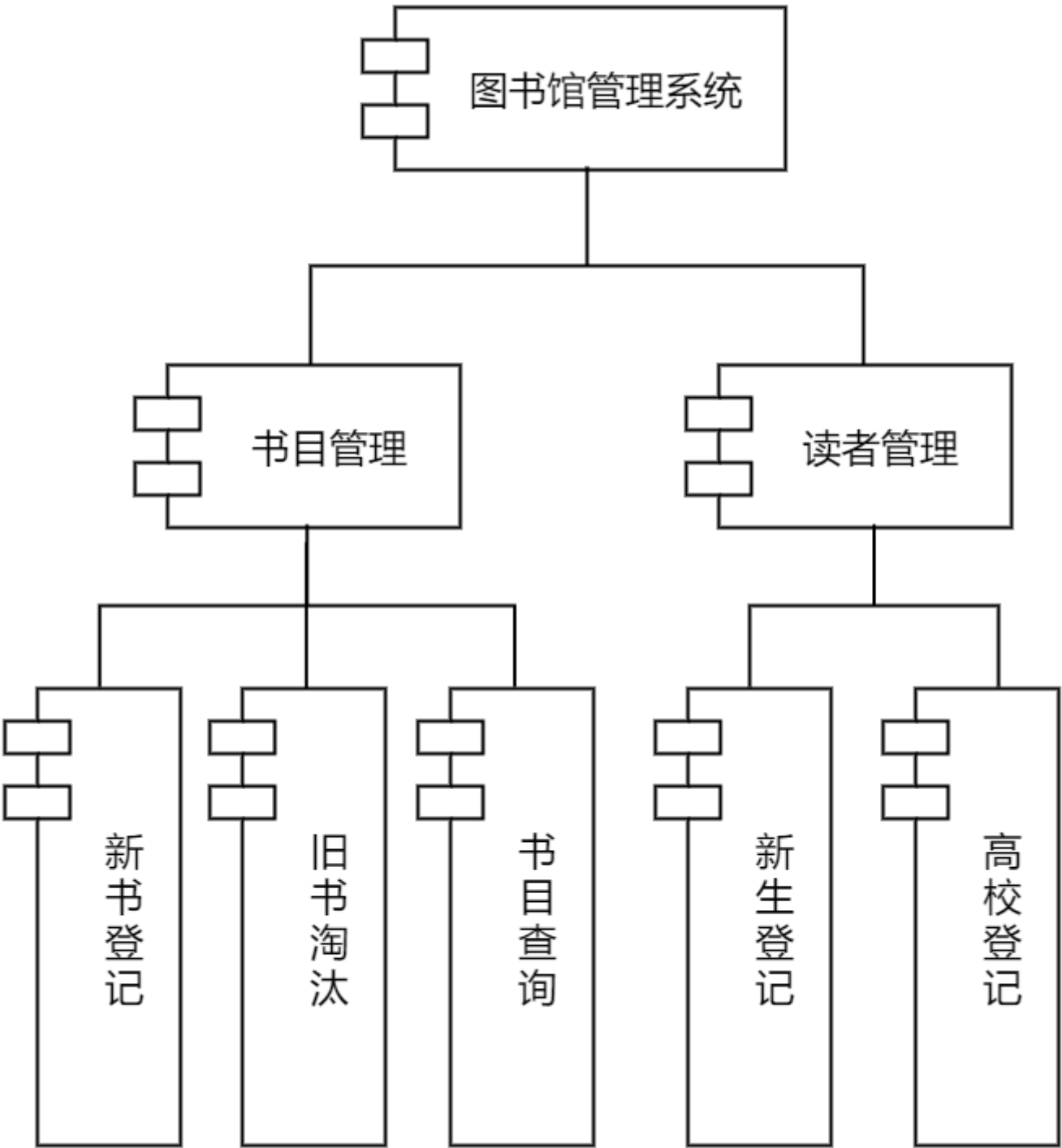
Experimental Procedure

Overall Design

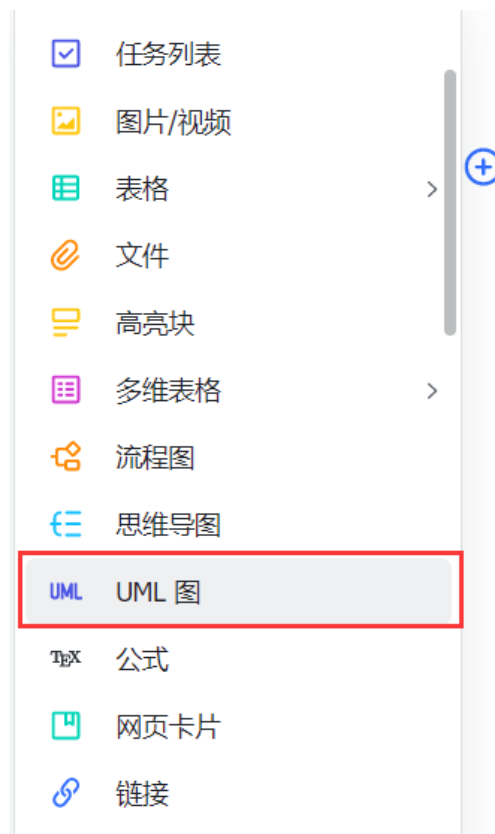
As a system developer, one needs to identify the most important concerns of a specific system and design a **general solution** to address these concerns. Architecture provides the highest-level design solution for a system. Therefore, we need to understand the architecture of a system to ensure that the system to be developed meets the expected characteristics.

Structure Diagram

Divide the entire system into parts, where the function of each part is simple and clear. Using functional decomposition can simplify the system structure to a certain extent, making the system easy to understand and modify.



It is recommended to add a "UML Diagram" in Feishu Docs to complete the drawing of the above structure diagram.



This is the [documentation tutorial](#) for drawing UML diagrams.

Design Category

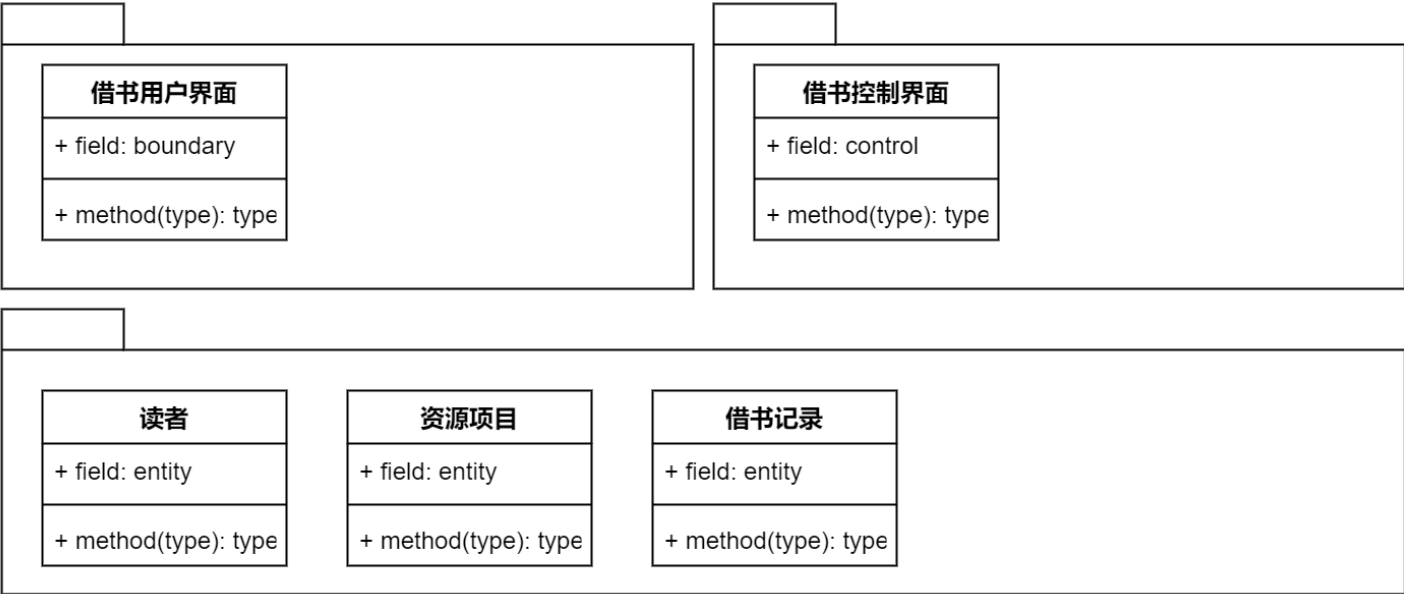
Modules in the system can be designed as **boundary classes, control classes, and entity classes**.

Boundary classes are used to describe the interaction between external actors and the system, and identifying boundary classes can help developers identify user requirements for the interface.

A control class is a class that controls the operation of other classes. Each use case typically has one control class, which controls the sequence of events within the use case, and control classes can also be shared among multiple use cases.

Entity classes are classes used to model information that must be stored and related behaviors. Entity objects (instances of entity classes) are used to save and update information about certain phenomena, such as events, people, or some real-life objects.

Please each development team identify and distinguish the categories to which each module in the system belongs, and list them accordingly. Here are some [learning materials](#).



Note that when a participant interacts with a use case, **it will surely lead to the derivation of a boundary class**. (For example, when "Librarian" (a participant) interacts with a use case "Borrow Books", it will lead to the derivation of a boundary class "Borrowing User Interface".)

The drawing method refers to the suggestions in the previous structural diagram.

Task Assignment

Since it is necessary to demonstrate teamwork, it is necessary to clarify the task division of each development group during the experiment. Meanwhile, dividing the tasks in the experiment according to status, priority, and responsible person, etc., will also help us complete the tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items. After creating a new Feishu Docs, you can copy the content from the template into your own cloud document.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group form (ebc9e17yfy.feishu.cn), **with the teaching assistant responsible for backup**.

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

Due to the limited space of the Feishu Free Edition, please try to **avoid directly uploading video files** to the group, so as not to affect other students' use of the upload-to-document function due to space limitations. It is recommended to use [Asen](#), a screen recording application, to record and automatically publish videos, and upload them to the Feishu group form ([ebc9e17yfy.feishu.cn](#)).

Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other (📅 **2025年4月23日 实验九**).
- **Before 11:15**, each group should complete the interactive evaluation ([ebc9e17yfy.feishu.cn](#)), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

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- Conduct experiments within the scope of legality and compliance to ensure personal safety
- The recommended software (tools) are suggestions and not mandatory requirements

Experiment 10 Activity Diagram and Sequence Diagram

Experimental Purpose

Continue to complete and write the system design report by learning to draw activity diagrams and sequence diagrams.

Experiment Introduction

This experiment continues with the third part of system development - system design. The main content of this hands-on experiment includes the drawing of activity diagrams and sequence diagrams.

Experimental Procedure

Define the task

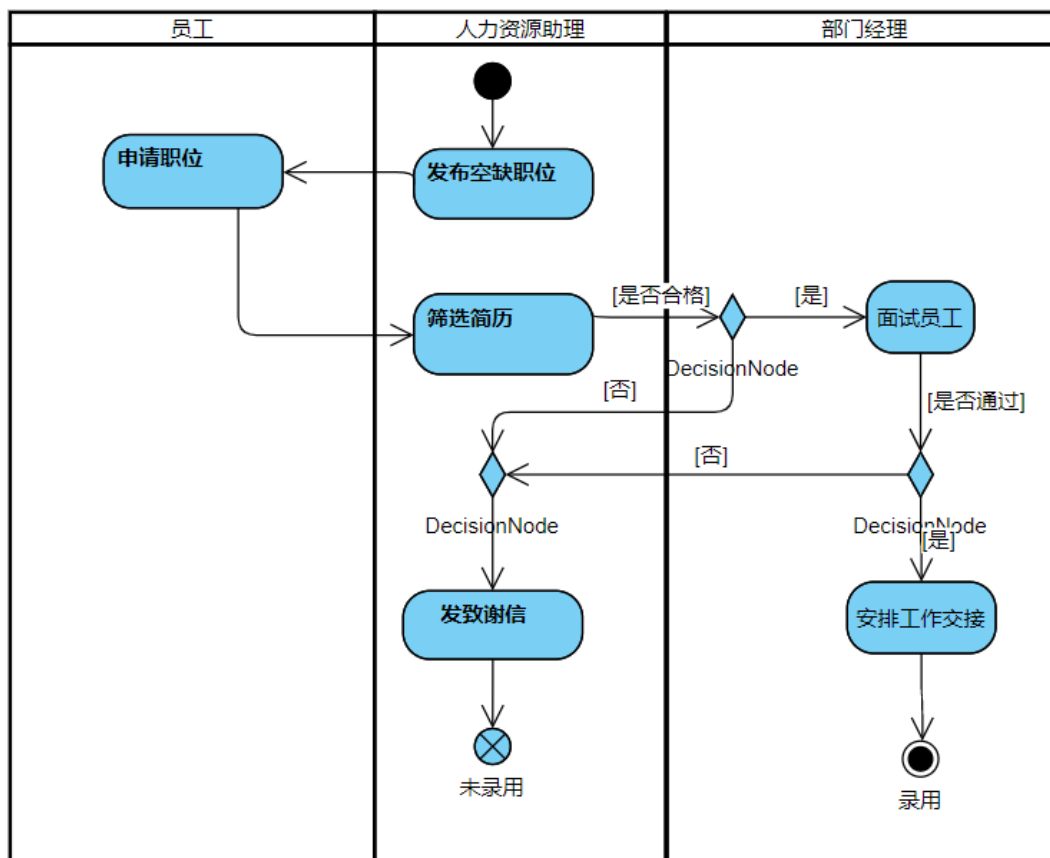
As system developers, we need to be proficient in the business workflow of the system and how various objects in the business interact with each other. The drawing of activity diagrams and sequence diagrams constructs the overall framework of the system for us, enabling the system to better meet user needs.

Activity Diagram

Activity diagrams are UML modeling specifications that use graphical representations to describe behavioral processes and can fully support various descriptions of processes in natural language. When complex behaviors are sequenced based on activity diagrams, analysts will have a clear train of thought, and designers will notice the impact of behavioral sequence on logical correctness and efficiency, which can greatly improve people's ability to understand and control behaviors.

Activity diagrams provide a visual description of activity processes and can be applied in systems, business, workflows, or other processes. These diagrams focus on the activities being performed and who (or what) is responsible for performing them.

Regarding the drawing of activity diagrams, we need to be familiar with the system's business workflow, mainly refining use cases, describing the sequence of activities, and presenting the control flow from one activity to another. Here are some [learning materials](#).



It is recommended to use[ProcessOn](#) to complete the drawing of the above activity diagram.

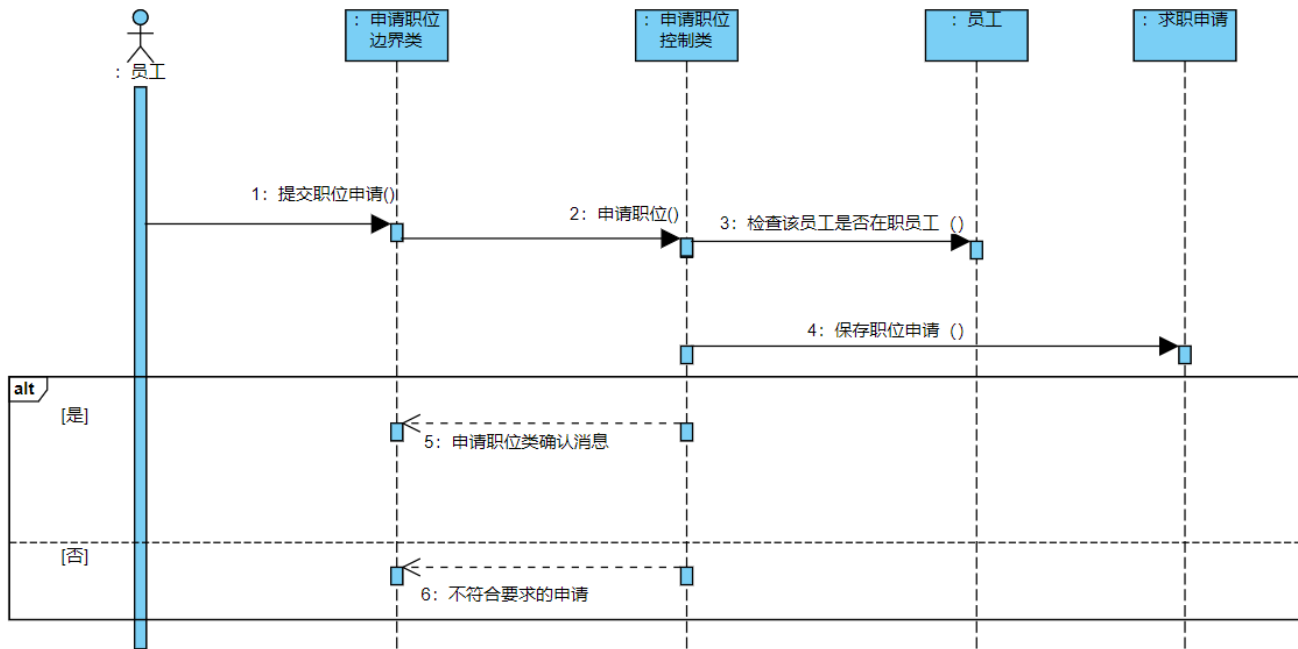
Here is the[User Manual](#).

Sequence Diagram

Sequence diagrams are created with use cases as units, examining which objects are required to complete the use case and how each object interacts with others.

By using sequence diagrams, we can clearly break down activities into interactions between objects, so that each behavior has a responsible party, which is a crucial step for both business process analysis and software process analysis.

Regarding the drawing of sequence diagrams, here are some[learning materials](#).



It is recommended to use [ProcessOn](#) to complete the drawing of the above sequence diagram. The user manual can be found in the previous activity diagram.

Note that for the drawing of the above two types of diagrams, each development team first needs to determine the number of diagrams to be drawn respectively (i.e., judge how many activity diagrams and sequence diagrams need to be drawn respectively based on the actual situation of the system). Secondly, considering the limited computer lab time, each development team is required to **complete at least two diagrams of each type, with two diagrams for each type**. If there is spare time, the remaining diagrams can be drawn together. Finally, if the number of diagrams to be drawn is more than two and time is insufficient, the drawing of the remaining diagrams will be supplemented after class.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.

任务看板



The above figure is an example, everyone needs to take screenshots of the initial task status and the final task status during each experiment and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

The Personal To-Do Kanban can be directly copied and pasted into your newly opened Feishu Docs.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions.**

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and upload it publicly via the Feishu group form (ebc9e17yfy.feishu.cn), **with the teaching assistant responsible for backup.**

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

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Time Node

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Precautions

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- The use of AI assistants is encouraged and promoted in this course, so you don't need to have any concerns. However, you must explain what issues or areas for improvement there are in the AI-generated results, and how you further communicated based on the preliminary answers provided by the AI. **Submitting AI-generated answers directly without review or modification is considered plagiarism** and will result in point deductions
- Conduct experiments within the scope of legality and compliance to ensure personal safety
- The recommended software (tools) are suggestions and not mandatory requirements

Experiment 11 System Detailed Design

Experimental Purpose

Write system design reports and proficiently master relevant system design methods.

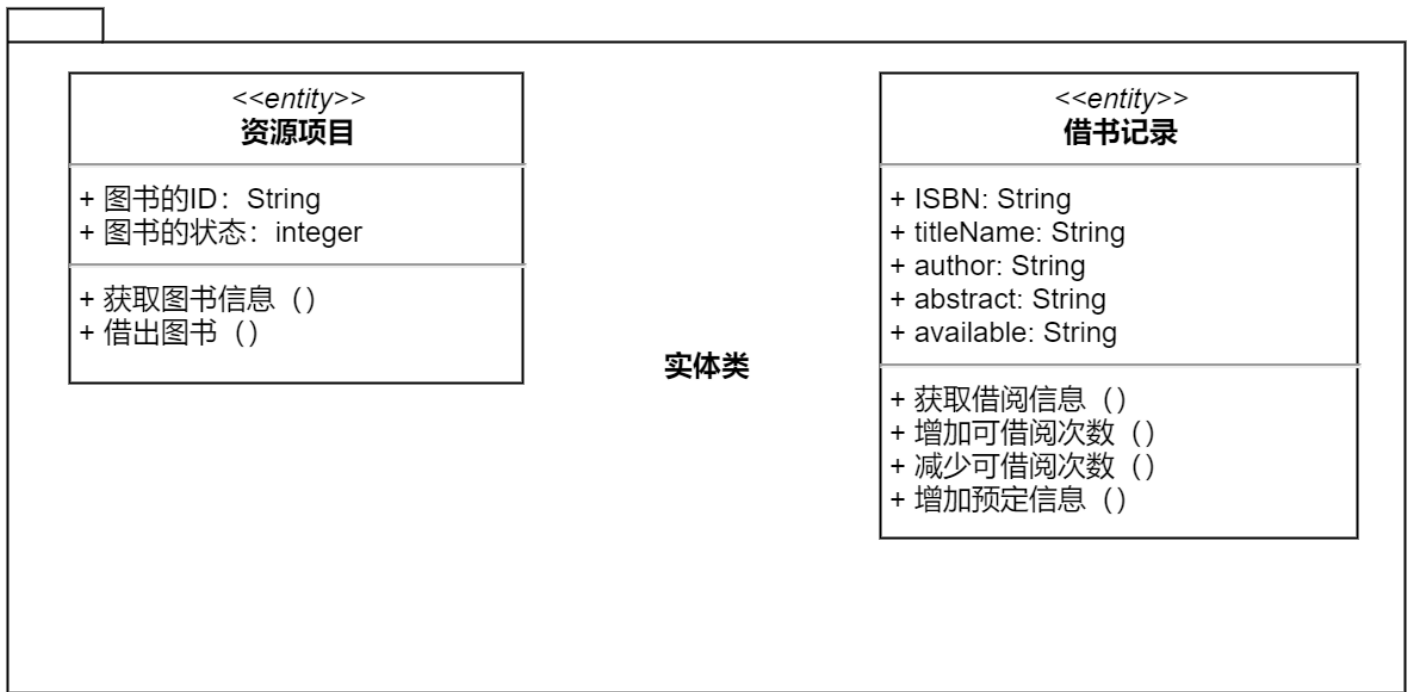
Experiment Introduction

This experiment continues with the third part of system development - system design. The main content of this hands-on experiment includes the detailed design of classes and the prototype design of the system.

Experimental Procedure

Detailed Design of Use Cases

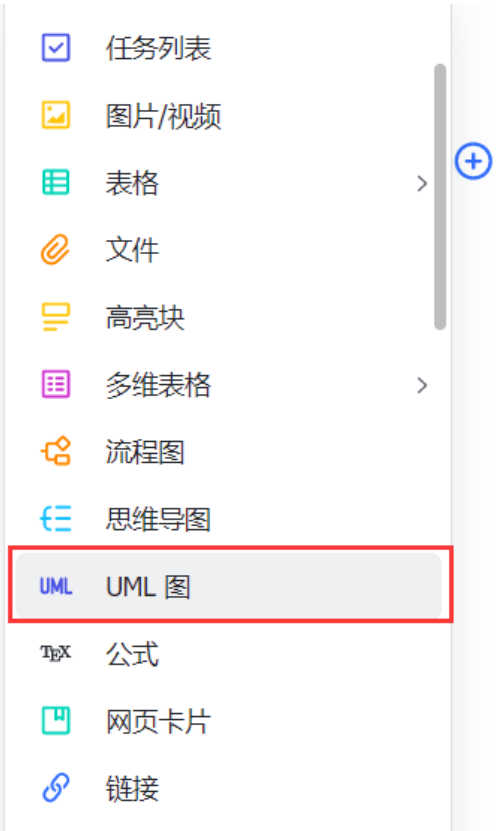
Through detailed design of classes, the system design plan can be further refined, making the entire model more concrete and facilitating the direct derivation of the programming design process.



Note that before designing classes in detail, we first need to identify the participating classes (boundary classes, control classes, and entity classes) of use cases, focusing on the message passing and its detailed definition in the sequence diagram. Each message in the sequence diagram can be mapped to an operation of a class and ultimately translated into a method of the class.

Here are the relevant [learning materials](#).

It is recommended to add a "UML diagram" in Feishu Docs to complete the detailed design of the above classes.

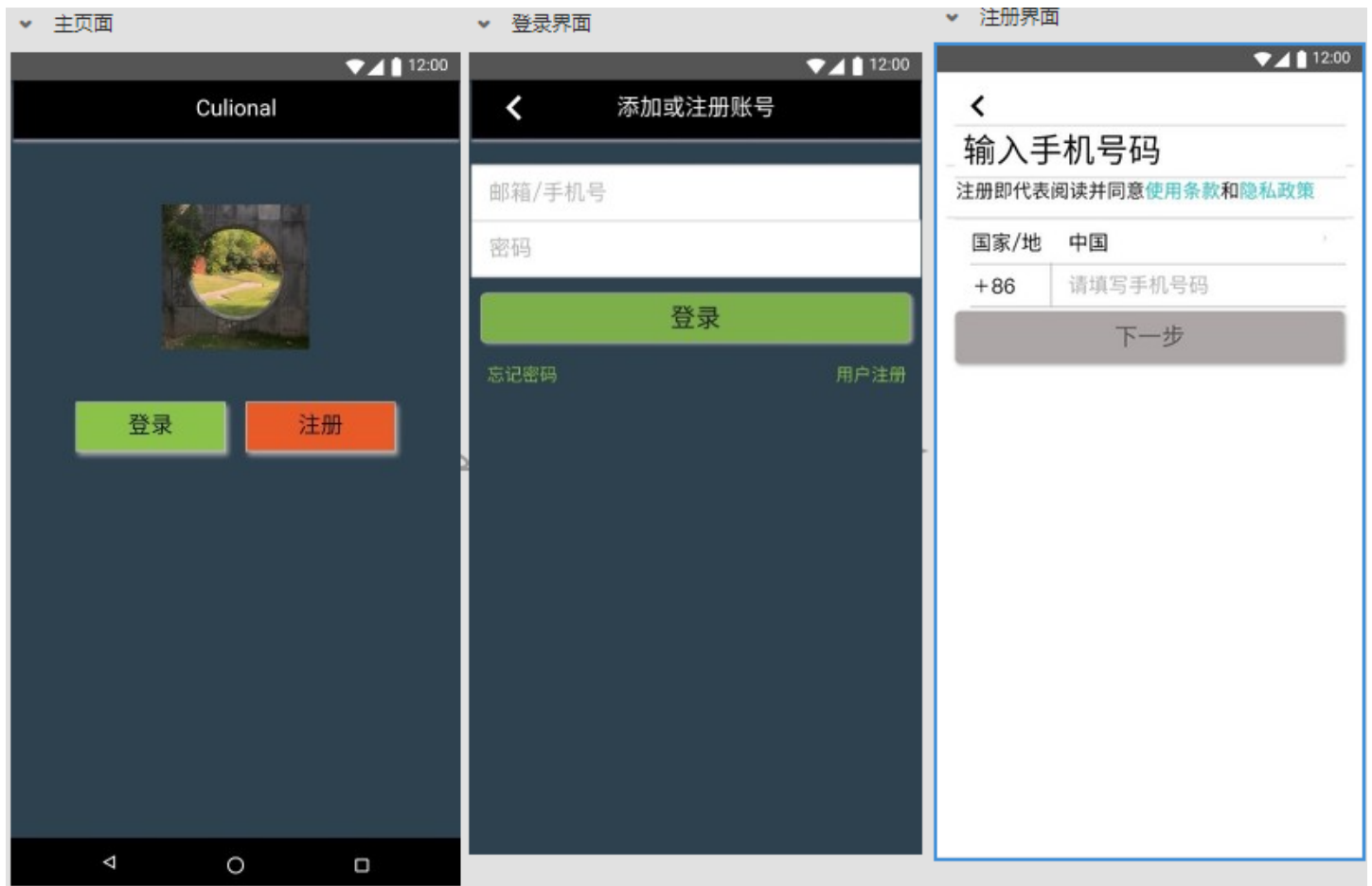


This is the [documentation tutorial](#) for drawing UML diagrams.

Prototype Design

System design often requires presenting abstract thinking in the form of prototypes, aiming to vividly represent ideas, functions, and content, thereby obtaining feedback and improving the product. Through prototype design, users can intuitively experience system elements and gain a preliminary understanding of the system's operations and functions.

This experiment will design the system login and registration interfaces.



Human-computer interaction is about designing, evaluating, and implementing interactive computer systems for people to use, mainly studying the information exchange between humans and computers. As system developers, we need to adhere to the user-centered principle, concisely and clearly reflect the characteristics of the designed system, thereby enhancing the user experience.

Note that the advantage of prototyping lies in its fast and concise characteristics. Therefore, everyone should focus on the key design points during the design process, and the refinement and beautification of the interface can be continuously improved in subsequent development.

It is recommended to use [Mockplus](#) to complete the design of the above interface.

Here is the introduction video of Mockplus RP [link](#), here is the official [text and image tutorial](#), here is the [beginner's operation reference tutorial](#).

Note: Previously, we noticed that some of your experimental reports only included pictures or had only one or two sentences of text description corresponding to the pictures, which could easily leave readers feeling confused. We hope that in future experimental reports, you can present a combination of pictures and text, with more detailed and comprehensive text descriptions.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group form (ebc9e17yfy.feishu.cn) , **with the teaching assistant responsible for backup** .

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

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Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年5月7日 实验十一](#)) .
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

Precautions

- Since there are many steps involved, please make sure to hurry up. Remember that "completion is more important than perfection". Do not overdo it by constantly making revisions, which may cause you to miss the submission deadline. Late submissions will be handled in accordance with the rules for late assignments.
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- Conduct experiments within the scope of legality and compliance to ensure personal safety
- The recommended software (tools) are suggestions and not mandatory requirements

Experiment 12 System Prototype Design

Experimental Purpose

Write system design reports and master relevant system prototype design methods.

Experiment Introduction

This experiment will continue with the third part of system development - system design. The main content of this hands-on experiment is the interface Interaction Design of the system's substantial functions.

Experimental Procedure

Prototype Design

System design often requires presenting abstract thinking in the form of prototypes, aiming to vividly showcase ideas, functions, and content, thereby obtaining feedback and improving the product. Through rapid prototyping, users can intuitively experience system elements and gain a preliminary understanding of the system's operations and functions.

The advantage of prototyping lies in its fast and concise characteristics. Therefore, during the design process, we should focus on the key design points, and the refinement and beautification of the interface can be continuously improved in subsequent development. Meanwhile, as system developers, we need to adhere to the user-centered principle, concisely and clearly reflect the characteristics of the designed system, thereby enhancing the user experience.

In the previous experiment, we had already completed the interface design for non-substantive functions such as registration and login. This experiment continues with the prototype design of the system, mainly focusing on completing the design of each substantive functional module of the system, and initially finalizing the system prototype.



It is recommended to use [Mockplus](#) to complete the design of the above interface.

Here is the introduction video of Mockplus RP [link](#) , here is the official [text and image tutorial](#) , here is the [beginner's operation reference tutorial](#) .

Considering the limited time available for hands - on practice, each development team needs to complete **the interface design of at least 4 substantial functions** . **Note that** the final presentation should not only be a static interface design but also include **dynamic interaction** between the user and the interface (which can be reflected in the recorded video explanation and the final system design report). If the number of functions to be designed exceeds 4, the content filling needs to be completed after class (before the next experiment).

- ✎ **In addition, after the completion of the main content of this experiment, it is necessary to review, analyze, and summarize whether there are any problems that were not discovered at the time in the report content prepared during the previous design phase, and list them in tabular form in this report.**

Each development team needs to submit the **overall system design report** via this form (ebc9e17yfy.feishu.cn) before the next hands-on experiment (May 21). Regarding the content of the report, it specifically includes: problem correction and content filling, with requirements the same as those in ([目 实验八 系统分析](#) : Overall Report of the System Analysis Phase).

If the time for this hands-on experiment is relatively ample, you can start writing the overall system design report, but it will not be used as a reference for the evaluation of this hands-on experiment.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and upload it publicly via the **form** (ebc9e17yfy.feishu.cn) in the Feishu group, **with the teaching assistant responsible for backup**.

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

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Time Node

- Experiment reports and videos, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other (📅 2025年5月14日 实验十二).
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
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Precautions

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Experiment 13: Drawing Code Flowcharts

Experimental Purpose

Write system implementation reports and proficiently master the production techniques of code flowcharts.

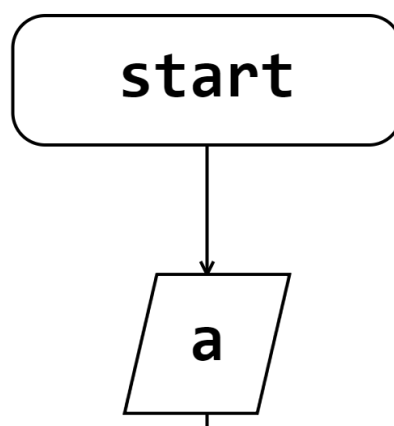
Experiment Introduction

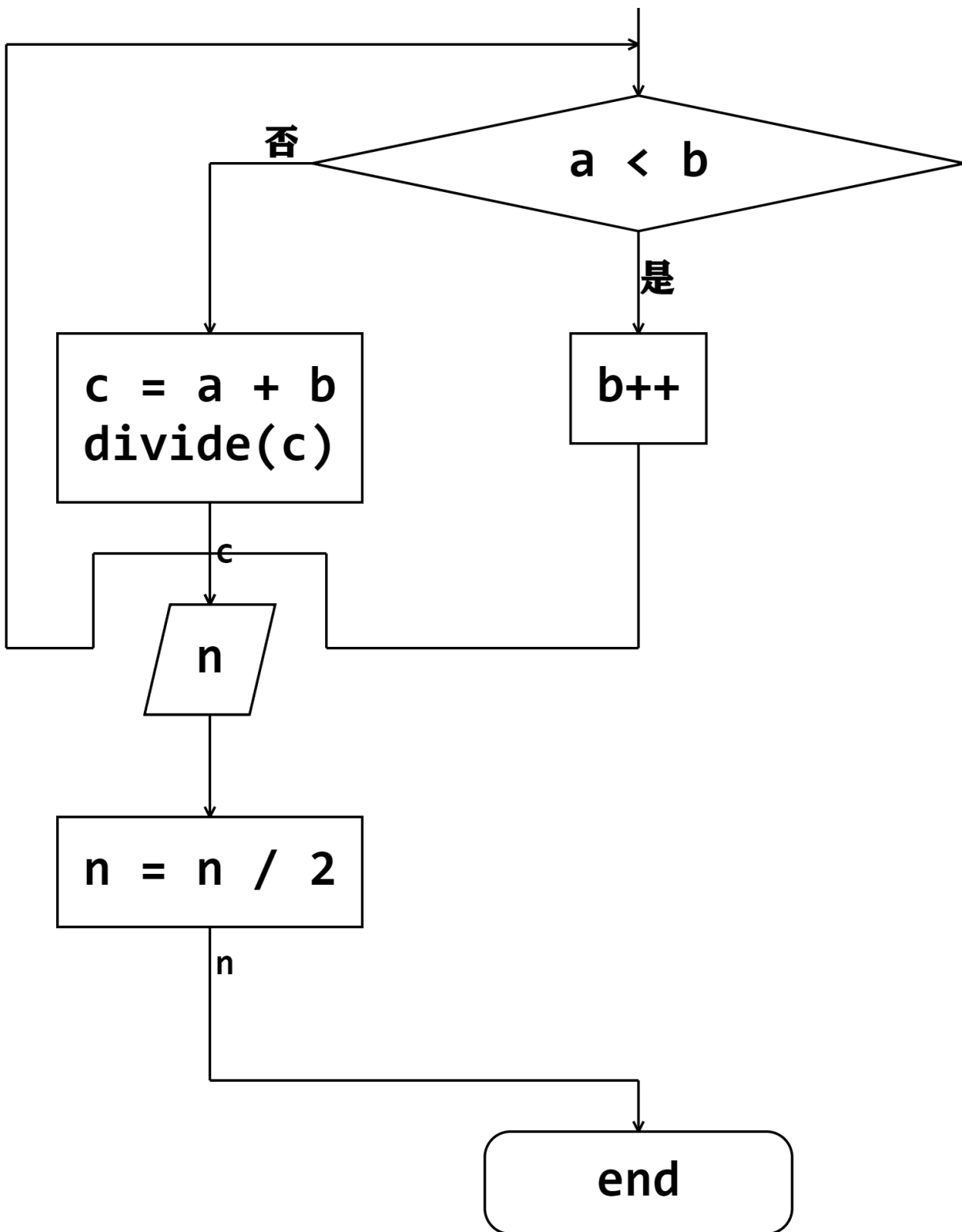
Last week, we completed the third part of the system development phase - system design. This experiment will commence the fourth part of system development - system implementation. The main content of this hands-on experiment is the drawing of code flowcharts.

Experimental Procedure

Code Flowchart

By drawing code flowcharts, it not only helps system developers efficiently clarify problems and master knowledge, but also facilitates self-discovery and problem-solving. Additionally, it aids developers in communicating and collaborating with other team members, making the communication process within the entire team smoother.





The above figure is a sample diagram. For this hands-on experiment, you need to draw the code flowcharts for the interfaces of all system functions (including non-substantive and substantive functions).

It is recommended to use [ProcessOn](#) to complete the drawing of the above code flowchart.

Here is the [User Manual](#).

Note that for the drawing of the above code flow, each development team first needs to determine the number of drawings (i.e., judge how many drawings are needed based on the actual situation of the system). Second, considering the limited computer time, each development team is required to complete the drawing of all non-substantive functions (registration, login, etc.) and at least 4 code flowcharts for substantive functions . Finally, if the number of drawings required is more than this, the drawing of the remaining flowcharts needs to be supplemented after class.

In addition, we need to further communicate with users and obtain feedback in the next experiment. Therefore, each development team should **contact the Target Users in advance** before the next experiment (May 27). At the same time, the system should also be basically implemented to allow users to try it out.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions** .

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group form (e9c9e17yfy.feishu.cn) , **with the teaching assistant responsible for backup** .

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Video Release

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Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年5月21日 实验十三](#)) .
- **Before 11:15**, each group should complete the interactive evaluation (e9c9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

Precautions

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Experiment 14: User System Prototype Interaction and Feedback

Experimental Purpose

Obtain feedback through users' actual interactions and communication, and write experimental reports.

Experiment Introduction

This experiment will continue with the fourth part of system development - system implementation. The main content of this hands-on experiment includes: designing user interview outlines, allowing users to actually operate the system, and further obtaining actual feedback after communicating with Target Users.

Experimental Procedure

Interview Preparation

In this experiment, each development team needs to summarize the results of each previous experiment, and on this basis, review and reflect on which functions of the system have been implemented and which have not been implemented so far. Based on these issues, write a corresponding interview outline and include it in the experiment report.

The content that the interview outline needs to include is:

1. Explain what operations users need to perform when trying out the system?
2. Describe how communication with users is conducted (online/offline), and provide detailed information such as time, location, and equipment.
3. Explain to users the specific operations and usage methods of the existing system functions.
4. Content of user communication:
 - ① Can the current system's functionality address the pain points and requirements originally raised by users?
 - ② What problems does the user think the current system functions have solved?
 - ③ Ask the user about their experience with the interface and functionality of the current system?
 - ④ Ask the user if there are any issues such as laggy or unsmooth performance during the system operation.
 - ⑤ The user raises issues and suggestions regarding the functions of the current system.
 - ⑥ Ask the user if any new pain points and demand issues have emerged after using the system?

Everyone should write the outline in combination with the actual situation of the system, and can continue to expand on the basis of the above content. Do not simply copy and paste.

In addition, before allowing users to actually operate the system, test the operating environment of the system, equipment, etc. within the group first to ensure the smooth progress of the entire interaction process. Here **it is required that everyone record the system debugging process (1. System Debugging Video)**, including the following content:

1. Demonstrate the system's non-substantive functions (such as registration and login) and all substantive functions.
2. During the demonstration, not only normal input values but also some abnormal values should be tested.

User Communication

Have the Target User actually operate the currently implemented system functions, record the entire process (to prove that it is a real user using it), and record their real feedback. Attach the recorded **video and text (2. User Practical Operation and Feedback Video)** to the end of the experimental report. (It is recommended to use the voice-to-text function of Feishu Meeting, and pay attention to setting viewing permissions when uploading the link.)

Based on communication and feedback from users, summarize the current progress and issues, and adjust the experimental plan according to the actual situation.

List the issues that users have proposed for improvement in tabular form. At the same time, list the improvement measures for these issues and set a deadline for each issue that needs improvement. After class, make modifications and implement them as planned.

In addition, **each development team needs to submit the overall system implementation report before the next experiment (6.4)**, the link is as follows: ebc9e17yfy.feishu.cn (the format and requirements of the report content refer to the overall report of the system analysis and design phase).

If there is sufficient time after completing the content of this experiment, you may start writing the overall system implementation report, but it will not be used as a criterion for this evaluation.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group (e9c9e17yfy.feishu.cn). **The teaching assistant is responsible for backup**.

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Record a short video to describe the key steps and analysis results of this experiment. The final video length **shall not exceed 3 minutes**.

Video Release

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Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年5月28日 实验十四](#)).
- **Before 11:15**, each group should complete the interactive evaluation (e9c9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
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- Conduct experiments within the scope of legality and compliance to ensure personal safety
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Experiment 15 System Testing

Experimental Purpose

Write system test reports and identify issues that occur during system operation.

Experiment Introduction

This experiment will commence the fifth part of system development - system operation and maintenance. The main content of this hands-on experiment is to initially complete the system operation test of the developed system through black-box and SAST white-box testing. In addition, each development team needs to conduct environmental testing via Feishu Meeting respectively and record the entire testing process.

Experimental Procedure

Test Process

The purpose of testing refers to executing a program in order to discover errors. A successful test is one that, through the design of a limited number of appropriate test cases, successfully discovers previously undetected errors.

SAST white-box testing

SAST white-box testing is a type of testing conducted on source code, and the main contents of the testing include lexical analysis, syntax analysis, static error analysis, and dynamic detection, etc. When designing test cases, the test object is regarded as a transparent box, allowing testers to utilize the internal logical structure of the program and its related information to design or select test cases, thereby testing the internal logical structure of the program.

Black box testing

Black box testing focuses on testing the functional requirements of software, and it detects whether each function can be used normally through testing. When using this method, testers completely ignore the internal logical structure and internal characteristics of the program, and only check whether the program's functions comply with its functional specifications based on the detailed description of the program module.

Here are some [learning materials](#) and [examples](#)

Note that, considering the limited time available for hands-on operation, each development team needs to complete at least two system functions for both black-box and SAST white-box testing. If there is more spare time, further improvements can be made to other system function tests.

Environmental testing

Within the Feishu group, you can share your screen by initiating a video meeting. During this operational test, each group must record the entire demonstration process (make sure to turn on the microphone and set the permissions for later sharing). The scope of the demonstration can be within your own group or include members from other groups, and the specific implementation will be decided by each group.



Note that since this experiment is only a preliminary test, it is recommended that everyone create a new Feishu group chat to initiate video meetings for sharing, so as to avoid chaos caused by everyone initiating meetings through the class Feishu group chat.



Conduct as many tests as possible. Through repeated testing and demonstrations from multiple aspects, we can identify some avoidable errors. During subsequent continuous improvements, each development team should also conduct as many tests as possible to prevent issues from occurring in the customer delivery phase of the next experimental session and the final end-of-term demonstration phase.

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



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Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and upload it publicly via the Feishu group form ebc9e17yfy.feishu.cn, **with the teaching assistant responsible for backup**.

See here for the general format structure.[Basic style of the Information System Practical Report](#)

Record Video

First, record a short video to describe the key steps and analysis results of this experiment. The final video length **should not exceed 3 minutes**.

Second, record the testing process for subsequent review.

Video Release

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Time Node

- Experiment reports and videos, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other (📅 2025年6月4日 实验十五).
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Precautions

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- The use of AI assistants is encouraged and promoted in this course, so you don't need to have any concerns. However, you must explain what issues or areas for improvement there are in the AI-generated results, and how you further communicated based on the preliminary answers provided by the AI. **Submitting AI-generated answers directly without review or modification is considered plagiarism** and will result in point deductions
- Conduct experiments within the scope of legality and compliance to ensure personal safety
- The recommended software (tools) are suggestions and not mandatory requirements

Experiment 16 System Delivery and User Feedback

Experimental Purpose

Deliver the complete system to the user, obtain feedback through the user's actual interaction and communication, and write an experimental report.

Experiment Introduction

This experiment will continue with the final part of system development - user delivery. The main content of this hands-on experiment includes: designing user interview outlines and presentation PPTs, introducing the overall system to users in detail, allowing users to actually operate the system, and further obtaining actual feedback after communicating with target users.

Experimental Procedure

System Testing

Before communicating with the user, complete all system testing and debugging to ensure that the user can operate normally.

Interview Preparation

During this experiment, each development team needs to summarize the results of each previous experiment, summarize the specific improvements made based on user feedback, and analyze whether the system can meet user needs. Based on these issues, write a corresponding interview outline and reflect it in the experiment report.

The content that the interview outline needs to include is:

1. Describe how communication with users is conducted (online/offline), and provide detailed information such as time, location, and equipment.

2. Inform the user of some key elements to note in system delivery (subject to the content of the delivery receipt).

3. Content of user communication:

- ① Whether the functions of the current system can address the pain points and needs originally raised by users.
- ② What improvements the development team has made in response to the opinions and suggestions put forward by users during the prototype communication phase.
- ③ Ask users about their experience with the interface and functions of the current system.
- ④ Ask users if there are any issues such as laggy or unsmooth operation during system runtime.
- ⑤ Users put forward issues and suggestions regarding the currently delivered system.
- ⑥ Ask users if any new pain points and needs have emerged after using the system.

Note: Everyone should write the outline in combination with the actual situation of the system, and can continue to expand on the basis of the above content, rather than simply copying and pasting blindly.

4. Fill out the delivery receipt based on communication with the user.

Delivery Receipt	
Delivery Time	
Delivery Object (Target User)	
Delivery Steps	
User Evaluation (Satisfaction)	
User Signature	

User Communication

Have the Target User actually operate the currently implemented system functions, record the entire process (to prove that it is a real user using the system), and record their real feedback. Attach the recorded **video and text** to the experimental report. (It is recommended to use the voice-to-text function of Feishu Meeting, and pay attention to setting viewing permissions when uploading the link.)

Based on communication and feedback from users, summarize the current progress and issues, and adjust the experimental plan according to the actual situation.

If users raise issues that need improvement, they shall be listed in tabular form. Meanwhile, improvement measures for these issues shall be listed, and a deadline shall be set for each issue that needs improvement. After class, modifications and implementation shall be carried out as planned.

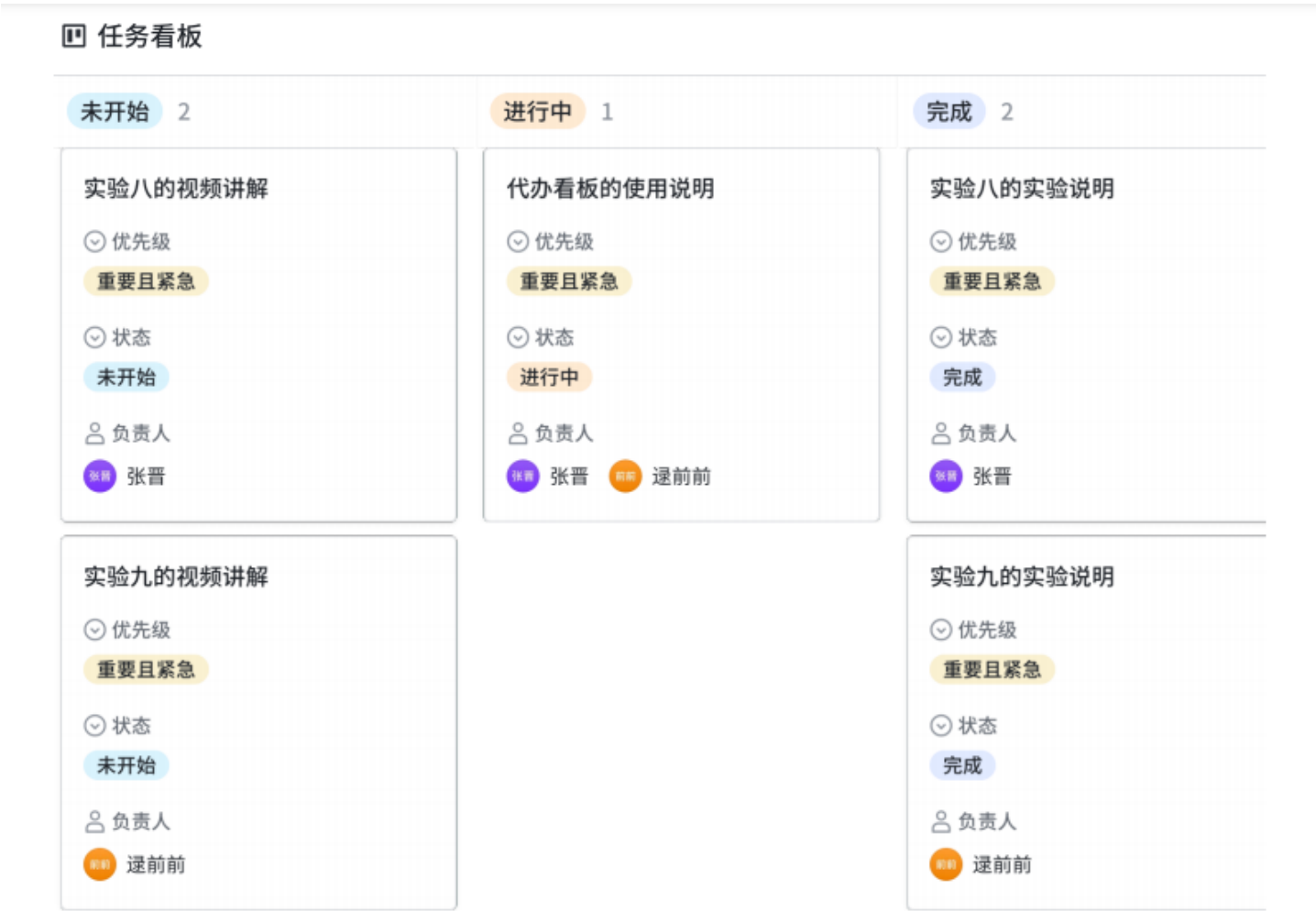
If there are many modifications, it is necessary to redeliver to the user after the modification and implementation are completed, and fill out the delivery receipt. Only the final delivery results and the delivery receipt can be included in the final course completion report.

The content that the video needs to include is:

- 1. Explain what operations users need to perform when trying out the system?
- 2. Explain to users the specific operations and usage methods of the existing system functions

Task Assignment

Dividing tasks in the experiment according to status, priority, and responsible person, etc., also helps us complete tasks in a more organized and efficient manner.



The above figure is an example. In each experiment, everyone needs to take screenshots of the initial task status and the final task status and include them in each experimental report.

It is recommended to use the [Personal To-Do Kanban](#) in the Feishu Docs Template Library to complete the assignment of the above to-do items.

Write a report

Write the experimental report in groups using text and images. Note that in the reports for each stage, not only should they include corresponding characteristic charts, **but also have matching text descriptions**.

The report is required to include at least one photo documenting the group discussion process. It is recommended to use Feishu Docs for collaborative writing to facilitate teamwork. **When submitting, please export it as a PDF file** and publicly upload it via the Feishu group ebc9e17yfy.feishu.cn. **The teaching assistant is responsible for backup**.

See here for the general format structure. [Basic style of the Information System Practical Report](#)

Record Video

Have the Target User actually operate the currently implemented system functions, record the entire process (to prove that it is a real user using the system), and record their real feedback.
Attach the recorded video and text to the experimental report. (It is recommended to use the voice-to-text function of Feishu Meeting, and pay attention to setting viewing permissions when uploading the link.)

Video Release

Since the free version of Feishu has limited storage space, please try to **avoid directly uploading video files** to the group to prevent space limitations from affecting other students' use of the upload-to-document feature. It is recommended to use [Asenscreen](#) recording app for video recording and automatic publishing.

Time Node

- The experimental report and video, **must be uploaded by 10:40** to allow sufficient time for students to review and evaluate each other ([📅 2025年6月11日 实验十六](#)).
- **Before 11:15**, each group should complete the interactive evaluation (ebc9e17yfy.feishu.cn), evaluating the other group's experimental report and video. And based on the specific division of labor, pose targeted questions to a particular classmate
- **Before 11:30**, the teaching assistant will review and release the grading results, and announce the excellent evaluation group by 20:00 that evening

Precautions

- Since there are many steps involved, please make sure to hurry up. Remember that "completion is more important than perfection". Do not overdo it by constantly making revisions, which may cause you to miss the submission deadline. Late submissions will be handled in accordance with the rules for late assignments.

- Team leaders should do a good job in division of labor and collaboration, fully mobilize the enthusiasm of team members, ensure timely completion, and guarantee the quality of the content.
- To protect privacy, the scope of publication for documents and videos **is restricted to the Feishu group of this course**, please **do not publicly disclose** or disseminate them.
- During offline computer-based sessions, if you have any questions, please ask the teacher directly; during online computer-based sessions, if you have questions related to the computer-based requirements and procedures, please **publicly ask** the teaching assistant in the Feishu group. In order to **ensure fairness**, private messages for help will not be answered in principle. **publicly ask** the teaching assistant in the Feishu group. In order to **ensure fairness**, private messages for help will not be answered in principle.
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Final Course Requirements for Information System Development Methods and Tools in 2025

Final presentation time

Lecture during the 17th teaching week (June 16, Monday, starting at 9:50).

Material Submission Time

Submit via the form and complete the peer review lottery by 20:00 on the evening of June 13 (the lottery will be distributed in the Feishu group at 19:00 on the same day). The submission link is as follows: e9c9e17yfy.feishu.cn

Complete peer reviews (targeted evaluation based on random draw) by 22:00 on the evening of June 13th. The submission link is as follows: e9c9e17yfy.feishu.cn

Material Submission Requirements

1. Course Completion Report (PDF format)

The report requirements are as follows:

[📄 Final Report Outline](#)

The report format is as follows:

[📄 \(Appendix 1\) Cover Page and Format Requirements for the Course Completion Paper of the School of Management, Tianjin Normal University](#)

Refer to the format requirements such as cover, font, indentation, numbering, etc. therein.

样例: [📄 Excellent Example of 21st Level System Development](#)

Note:

Sample **for reference only**, must be strictly **completed according to the requirements of this year**.

2. System Code Related Materials

Uniformly upload a Compressed Packet containing source code, resources, and operating instructions. If you can provide the corresponding online operating environment and ensure its normal operation, appropriate bonus points will be awarded.

3. Video Recordings (2): All submitted in the form of links

- System delivery process: It is required to be real and effective. There should be detailed recordings of the actual hands-on process of users.
- System Demonstration Process: The demonstration requires a detailed explanation of the overall system, presentation of **core code**, and demonstration of **key functions** in operation.

4. Delivery Receipt (showing delivery time, target user, delivery steps, satisfaction, and user signature)

It is necessary to privately submit the contact information of the Target User for subsequent random inspections. If the user cannot be reached after three attempts, **the score for the final delivery part will be 0**, and according to the course rules, **all subsequent bonus points will be cancelled**. Moreover, **an investigation will be launched**. If other issues such as plagiarism, piracy, and fraud are found in system analysis, design, and implementation, liability will be pursued.

Delivery Receipt	
Delivery Time	

Delivery Object (Target User)	
Delivery Steps	
User Evaluation (Satisfaction)	
User Signature	

5. On-site presentation during class

Slides are not allowed, use whiteboard + handwriting (same as the usual requirements for excellence evaluation)

Describe the advantages, demonstrate in-depth thinking and design, with a presentation time of within 3 minutes, and strictly prohibit exceeding the time limit.

The entire presentation process will be recorded by the teaching assistant.

After the presentation, there may be a Q&A and suggestion session with teachers.

Composition of Final Grades

Basic Score

- System that can run correctly: 20 points
- Complete Development Document: 20 points
- Successful Delivery: 20 points
- Peer Review: 10 points (affects the score of the evaluating group, not the score of the evaluated group)

The above scores (70 points in total) shall serve as the group's basic score

Bonus Points

- Sign up for the merit evaluation. Teaching assistants will select 10 groups to participate in the final merit evaluation presentation, randomly select 2 other groups for review. The order of the merit evaluation presentation: first the merit evaluation groups, then the on-site randomly selected review groups
- For groups that pass the evaluation, each group's final grade will be increased by 10 points
- After the evaluation and review, the whole class will vote (3 votes for the teacher, 1 vote for the teaching assistant, and 1 vote for each group) to select the top groups (6 groups) at the end of the semester. Each top group will receive an additional 10 points (cumulative 20 points) in their final grades.

- The group with the highest number of votes will be the final winning group, and the final grade of the winning group will be further increased by 5 points (cumulative 25 points).
- When a tie occurs in the vote, a targeted revote will be held.

Role Bonus

- Each time a weekday computer task is completed, the team leader of the development group will receive an additional 1 point in their individual final grade
- Each group that passes and is rated excellent in the regular computer lab experiments will receive an additional 3 points for their final course grade
- For each lecture, the student with the highest total score will receive an additional 3 points added to their final grade.
- Class representative gets +20 points for final exam score

Note:

- Only development teams that have successfully delivered can receive additional scores
- At the end of the term, all grades will not have decimals and will be rounded off for calculation.
- Those with a final total score exceeding 100 points shall all be calculated as 100 points and cannot be added to other items such as regular performance or computer-based test scores.

Final Demonstration Classroom Procedure

Click on the image to view the full table

Note:

- This is for viewing materials of each group during the evaluation [📄 2025年信息系统开发期末材料提交与互评](#) .
- The final report must be submitted in hard copy (to be collected by the end of class on June 16).